Enhancing CubeSat Computer Vision: A YOLOv8 Fine-Tuning approach for Real-World Celestial Detection

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Abstract. This study presents the development and optimization of a computer vision model based on YOLOv8 for the detection of celestial bodies, specifically the Moon, in imagery captured for a CubeSat aerospace mission. The work focuses on fine-tuning the YOLOv8 architecture to maximize detection precision under real-world conditions using a custom dataset composed of 8000 images. A detailed training methodology is applied, including hyperparameter tuning and analysis of performance metrics such as mAP@50. Final training results achieved a peak performance of 91.23% mAP@50. The impact of hyperparameters is examined in depth.

Keywords- Computer Vision, Machine Learning, Cubesat

1. INTRODUCTION AND JUSTIFICATION

As of now, nanosatellites have become the backbone of various industries, playing a critical role in communication, weather monitoring, navigation, Earth observation, and aerospace research [1]. Their small size, relatively low cost, and rapid deployment capabilities have revolutionized access to space-based services, enabling not only governments and large corporations but also universities, startups, and even high schools to participate in space exploration and research [2]. This democratization of space has led to an explosion in the number of missions launched, particularly in low Earth orbit (LEO), where nanosatellites are often used for high-frequency data collection and real-time monitoring.

Thanks to standardized platforms such as CubeSats, the design and development process has become more accessible and modular, significantly reducing both time and financial barriers. These small satellites are now being utilized for tasks that once required large, expensive spacecraft—from environmental monitoring and disaster response to technology testing and intersatellite communication. As launch opportunities grow and commercial satellite constellations expand, the demand for smarter, more autonomous nanosatellites is increasing. Yet, despite their growing presence and potential, a significant limitation remains: the lack of proven flight missions that incorporate artificial intelligence (AI) in their control algorithms. This gap in innovation severely constrains the range of possibilities considered by designers and engineers. As a result, mission planners are often forced to rely on conventional, less adaptive systems, leading to increased development costs, reduced operational flexibility, and a narrower scope of achievable objectives. Without integrating advanced AI capabilities into nanosatellite missions, the full potential of these technologies remains untapped, ultimately decreasing their reach and long-term impact across industries.

2. Problem Description

The Cubesat-02 project involves the structuring of a microsatellite that enables precise observation and photography of the moon from the upper atmosphere. Fundamentally, aims to explore the advantages offered by neural networks for precise image adjustments and the generation of control algorithms for out of orbit applications.

This documentation focuses on a comprehensive data analysis of the training process of the YOLOv8 model, employed for object detection within the captured imagery. The primary goal is to optimize the model's performance metrics, specifically the mAP-50, to enhance detection accuracy and reliability, critical factors for the mission's success. Methodologies, results, and recommendations for continuous model improvement are presented to ensure the model meets the demands of the flight mission and contributes effectively to the project's overall objectives.

3. State of the art

A small satellite is generally considered to be any satellite that weighs less than 300 kg. A CubeSat, however, must conform to specific criteria that control factors such as its shape, size, and weight [1]. These design guidelines are specified in a document provided by Cubesat initiative founder university called "Cubesat Design Specification".A CubeSat is a class of satellite that adopts a standard size and form factor, whose unit is defined as 'U'. A 1U CubeSat is a 10 cm cube with a mass of up to 2 kg [2]. larger sizes have become popular, such as the 1.5U, 2U, 3U, 6U, and 12U [1].

The CubeSat Project began as a collaborative effort between Prof. Jordi Puig-Suari at California Polytechnic State University (Cal Poly), San Luis Obispo, and Prof. Bob Twiggs at Stanford University's Space Systems Development Laboratory (SSDL) in 1999. The intent of the CubeSat Project was to reduce cost and development time, increase accessibility to space, and sustain frequent launches [2].

Similarly, the YOLO (You Only Look Once) architecture transformed computer vision by offering fast and accurate object detection in a single pass [3]. YOLOv8, released in 2023 by Ultralytics, represents the latest advancement with improved real-time performance and extended capabilities like instance segmentation. Its efficiency makes it suitable for edge devices, including use aboard CubeSats for on-orbit image processing. Together, these compact space platforms and optimized AI models are unlocking powerful new capabilities such as autonomous satellite inspection and real-time Earth observation, highlighting how convergence in hardware and software is reshaping both industries [4].

4. CubeRT-02

4.1. Exploratory Data Analysis

The data collection began with an initial set of approximately 800 images sourced from Google Images. While this served as a valid starting point for addressing a bibliography problem, it was not sufficient for real-world application. Given that the model is intended for deployment in an aerospace context, it became essential to integrate data that more closely reflects the actual detections the model will need to perform in the field.

To address this, a dataset of 10,000 images was captured over a six-month period using local equipment such as cellphones and amateur cameras. This collection was deliberately designed to encompass a wide range of scales and perspectives of the object of interest, thereby providing more representative and diverse data for the model to learn from in real-world aerospace scenarios.

A cleansing phase was then conducted to remove any blurry or poor-quality images from the locally sourced dataset, as well as to exclude any irrelevant or low-quality results obtained from web scraping. Image annotation was finally performed using Roboflow platform, additionally implementing a data augmentation technique to further enhance the dataset.

4.2. Model Selection

YOLOv8 is a neural network architecture designed for object detection, representing a fast and accurate solution for computer vision applied to object detection [5]. The model is composed of three principal components: the Backbone, the Neck, and the Head, each fulfilling distinct yet complementary roles in the object detection pipeline.

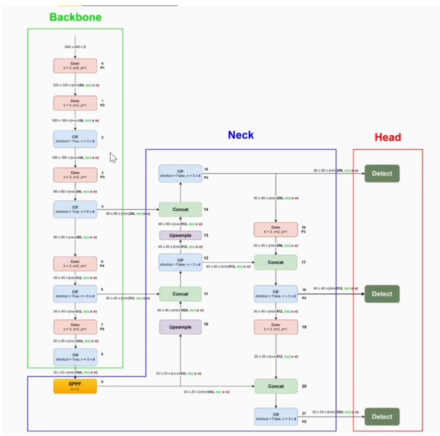

Figure 1. YOLOv8 overall architecture

The Backbone serves as the primary feature extractor. It processes the input image through a series of convolutional layers designed to capture hierarchical representations of visual features, from low-level edges to high-level semantic information.

The Neck component aggregates and fuses feature maps obtained from multiple scales within the Backbone. This multi-scale feature fusion is crucial for detecting objects of varying sizes.

Finally, the Head performs the detection task by predicting bounding boxes, objectness scores, and class probabilities based on the fused features from the Neck.

4.3. Model Training

The training process aims to optimize the hyperparameter configuration for YOLOv8, considering the performance metrics obtained after each training session. The primary focus is on maximizing the mAP50 metric, which is used to assess the model's precision and effectiveness in moon detection task.

The standard training procedure consists of a 15-epoch process, utilizing a random 20% subset of the total moon dataset for each iteration. Following this initial phase, an iterative training process incorporating hyperparameter tuning was carried out. A detailed analysis of the training process can be found in the appendices section. Optimal model performance was achieved with a learning rate in the range of 0.1 to 0.001, a momentum value of 0.98, and L2 regularization (weight decay) set to 0.001. Notably, no data augmentation techniques were employed, as their application was found to reduce model precision based on performance analysis. Based on the optimal hyperparameter configuration, a final training phase was conducted using the entire dataset over 100 epochs to maximize model performance.

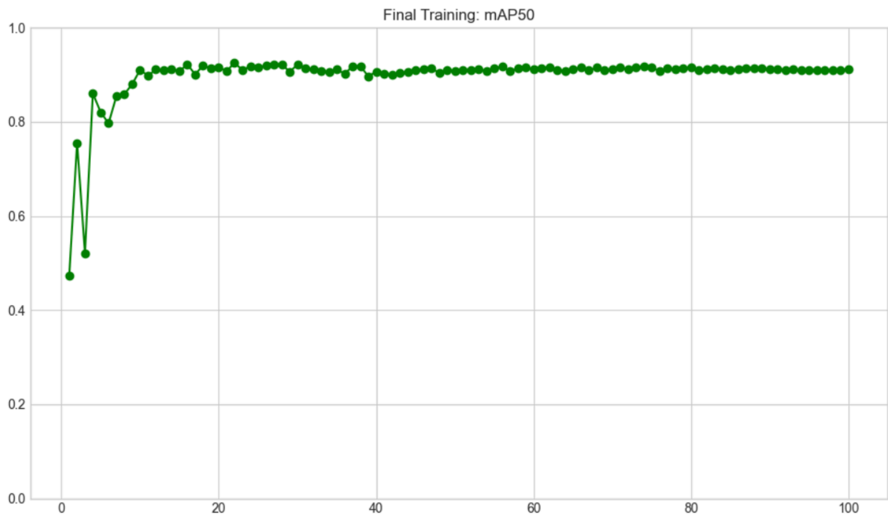

Figure 2. YOLOv8 Final Training. X axis: epochs. Y axis: mAP50

5. Discussion

The final mAP@50 achieved with the current dataset and YOLOv8 configuration was 91.23%. Data augmentation parameters did not contribute positively to the training process. This is likely due to the dataset already incorporating augmentation techniques, as noted in the Exploratory Data Analysis section. The presence of pre-applied augmentations may conflict with YOLOv8's internal augmentation pipeline, potentially resulting in excessive or redundant image transformations. Such over-modification can reduce the diversity and representativeness of the training data, ultimately leading to suboptimal model performance.

The performance data analysis could be further enhanced by incorporating a comprehensive testing phase involving the entire dataset. This is particularly relevant given that several hyperparameters—such as momentum, weight decay, and the final learning rate—play a significant role during the later stages of the training process. A subsequent training phase, informed by these insights, could leverage the refined hyperparameter configuration to enable more accurate and efficient learning, building upon the foundational results presented in this study.

6. References

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7. Appendices

Training	1	2	3	4	5	6	7	8	9	10	11	12	13	14
lr0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
lrf	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
momentum	0.937	0.937	0.6	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98
Weight decay	0.0005	0.0005	0.0	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Warmup ephocs	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Warmup momentum	0.8	0.8	0.8	0.8	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98
Degrees	0.0	0.0	0.0	0.0	0.0	0.0	0.225	0.45	0.0	0.0	0.0	0.0	0.0	0.0
Translate	0.1	0.1	0.1	0.1	0.1	0.0	0.45	0.9	0.0	0.0	0.0	0.0	0.0	0.0
Scale	0.5	0.5	0.5	0.5	0.5	0.0	0.45	0.9	0.0	0.0	0.0	0.0	0.0	0.0
Flipud	0.0	0.0	0.0	0.0	0.0	0.0	0.5	1.0	0.0	0.0	0.0	0.0	0.0	0.0
Fliplr	0.5	0.5	0.5	0.5	0.5	0.0	0.5	1.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	5.0	10.0	0.0	0.0	0.0
Perspective	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0005	10.0	0.0	0.0	0.0
Mosaic	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.0	0.5	1.0	0.0	0.0	0.0
Mixup	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	1.0	0.0	0.0	0.0
hsv_h	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.0	0.05	0.1
hsv_s	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.0	0.45	0.9
hsv_v	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.0	0.45	0.9
Copy_paste	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	1.0
Results (%)														
mAP50	53.2	58.4	60.2	60.2	60.2	53.2	49.1	32.3	68.9	64.4	68.6	70.5	63.5	60.8
mAP50-95	19.4	19.8	20.7	19.6	19.6	19.4	15.4	09.3	24.9	21.2	22.0	24.3	22.5	23.0